Research Proposal

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AI-Driven Procedural Content Generation for Construction Safety Training

1. Problem, Limitations, and Research Gap

1.1 Motivation

Construction hazards are both dynamic and spatial: workers need to be aware of risks in three-dimensional space throughout a job, under time pressure, and with moving machinery and co-workers in close proximity. Immersive VR/AR training allows for the exposure of trainees to such risky situations without risk. Recently reviewed studies show more positive outcomes from immersive education compared with conventional lecture or video-based training [1]. The main bottleneck, however, is not visual rendering but the time and effort required to develop training content.

1.2 Limitations of Current Practice

Manually developed virtual safety-training environments and current industry training approaches share five recurring limitations:

- Developing a virtual training scenario requires developers to manually place equipment, define hazards, configure interactions, and adjust the scene according to safety requirements. Building multiple scenarios for different training situations therefore requires considerable development time.
- A training environment created for one site or construction phase cannot always be reused for another. Changes in floor plans, equipment, trades, and ongoing activities often require substantial modification of the original scene.
- Most systems present trainees with predefined hazards and interactions. After repeated exposure, trainees may begin to remember where hazards appear rather than actively identifying them from the surrounding conditions. The same level of difficulty may also be unsuitable for both beginners and experienced workers.
- Hazards represented in a manually created environment are not necessarily linked to a particular OSHA requirement, project hazard record, or previous accident or near-miss. This makes it harder to verify why a hazard was included and whether the scenario reflects current safety knowledge.
- Language-guided 3D generation and PCGML techniques can reduce some of the effort involved in creating virtual environments [2]. However, generating a plausible scene is different from generating a valid safety-training scenario. Additional mechanisms are needed to check spatial constraints, regulations, and the intended learning objectives before generated content can be used for training.

1.3 Research Gap

The previous research has shown that (a) PCGML can be utilized to create functional designs that work [3], (b) the process of constructing a 3D scene can be made sense of using language processing technology [2], (c) a formal, ontology-grounded schema has been developed for BIM- and sensor-driven virtual safety training [4], and (d) AI has been used to personalize safety instruction [5]. However, no existing system connects these techniques into a single safety-oriented framework for construction training. A safety-critical generator must not only produce a visually plausible scene but also preserve task logic, spatial feasibility, regulation-linked hazards, and instructional intent, while exposing why a generated condition was accepted or rejected.

2. Research Questions and Innovation

- ***RQ1 -*** How can prompts, BIM/drawings, safety requirements, accident knowledge, and trainee data be fused into a common machine-readable representation for generation?
- ***RQ2 -*** Can a hybrid PCGML + constraint-validation pipeline generates diverse scenarios with ≥90% expert-rated hazard correctness while cutting manual authoring effort by ≥60%?
- ***RQ3 -*** Does adaptive, generated training improve hazard-recognition accuracy and response time compared with static, manually built virtual scenarios?

Scientific novelty. A Construction Safety Intermediate Scene Representation (CS-ISR) will explicitly encode objects, tasks, spatial relations, hazard states, required controls, regulation provenance, learning objective, and difficulty. This is an auditable bridge between AI reasoning and 3D execution that current PCG-for-training work does not provide. This differs from the ontology-grounded schema of Speiser and Teizer [4] in scope and function: their framework formalizes BIM- and sensor-driven hazard simulation for a defined site, whereas CS-ISR is designed as a generation-time intermediate representation that also encodes learning objective, difficulty, and generation provenance, so it can drive procedural variation rather than only structure a fixed scenario.

Technical novelty. A large language model is used as a grounded semantic planner instead of an unlimited scene creator. In fact, the developed PCGML allows for variation, whereas the deterministic geometrical and safety regulations provide elimination and correction of unsafe results. This makes it possible to use the controllable and relation-first logic exhibited in language-based scene generation[2] in the area where correctness plays significant importance compared to mere plausibility.

3. PCGML Framework

Representation: Here, content is represented not as pixels or mesh but as an object/relationship graph; for example, "worker-near-opening" or "forklift-crosses-pedestrian-route". Each node stores its class, dimensions, task role, hazard state, and configuration variables.

Generation: A conditional transformer/graph model learns distributions of legitimate object, activity, and hazard configurations from expert scenarios, BIM-derived cases, and reconstructed incidents. PCGML rightly fits the situation as the purpose here is functional and interactive knowledge gained from experience and not static writing of rules [1].

Controlled output: Generation produces multiple candidate CS-ISR scene graphs. Unity assets are not developed by mapping the semantic elements to a carefully selected library of prefabs, thus allowing the research effort to concentrate on creating and validating content and facilitating its adaptation as opposed to building 3D assets.

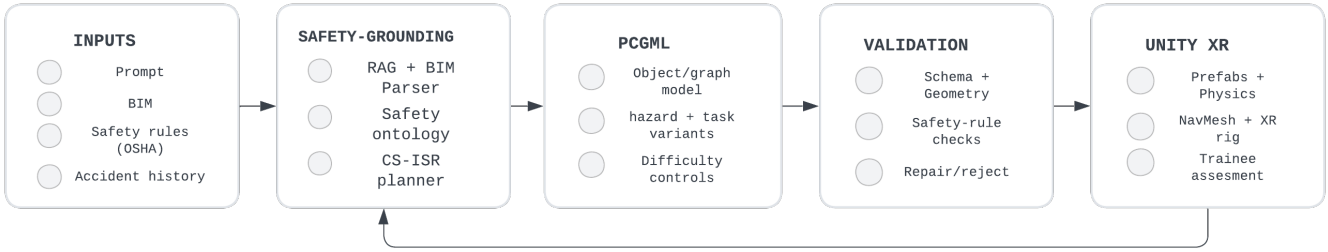


Figure 1. End-to-end PCGML pipeline - inputs are planned into a CS-ISR, generated, validated/repaired, and instantiated in Unity XR; trainee performance closes the loop back into scenario grounding.

4. Input and Knowledge Sources

Table 1 summarizes how each information source is grounded into the CS-ISR representation before generation begins.

Source	What it Contributes	CS-ISR Output
Prompt / learning goal	Trade, task, hazard focus, difficulty level	Training objective + scenario intent
BIM	Geometry, floors, openings, work zones, anchors	Scene graph + valid placement regions
OSHA / project safety rules	Required controls and prohibited states	Machine-checkable constraints + provenance
Accident / near-miss records	Real hazard combinations and causal context	Hazard priors + scenario templates
Trainee performance history	Experience level, missed hazards, response time	Difficulty tuning + personalized hazard selection

5. Content Validation

The following thresholds are prototype-phase design targets rather than benchmarks drawn from prior systems; they define what "acceptable" output means for this pipeline and will be revisited once early results are available.

- ***Structural validity -*** ≥95 of 100 generated outputs conform to the CS-ISR schema before scene creation begins.
- ***Geometry and physics validity -*** ≥98 of 100 placements collision-free and ≥95 of 100 required task locations reachable via NavMesh/path checks.
- ***Safety-rule validity -*** No more than 1 in 20 hazards requires manual correction after automatic repair; every accepted hazard carries rule/control provenance. BIM-based hazard checking supports the feasibility of this layer[6].
- ***Expert validity -*** Target mean rating of at least 4 out of 5 for realism and instructional usefulness, and at least 9 out of 10 hazard/control labels agreeing with expert consensus.

6. Personalization and Adaptation

Learner model: Unity logs identifies the hazards, unsafe actions, successful tasks, time taken to detect hazards, hints usage, as well as the repeated mistakes. A compact mastery profile is updated after every session.

Adaptation policy: In initial trials, simple techniques are applied – novices are shown slightly less complicated and more obvious hazards and some advanced users experience more subtle threats, simultaneous risks along with time pressure. In later phases, when there is a great deal of experience data available, contextual-bandit methods may be applied.

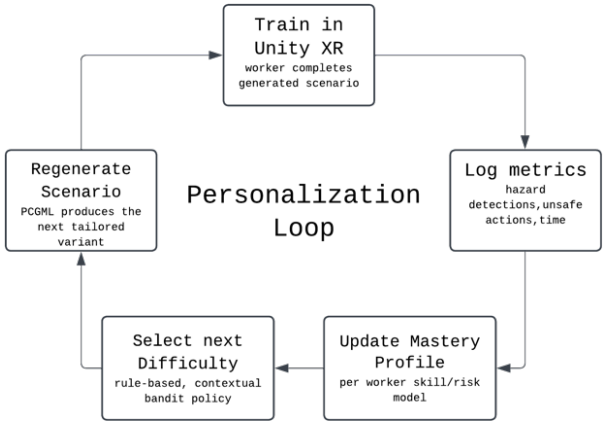


Figure 2. Closed personalization loop - training performance updates a profile that selects and regenerates the next scenario's difficulty.

7. Implementation Architecture

Table 2 maps each layer of the proposed system to concrete tools and measurable performance targets.

Layer	Technology	Role	Target
AI / ML	Python, PyTorch	RAG, CS-ISR planner, PCGML generator, adaptation policy	Scenario generated ≤ 10 s
BIM	IFCOpenShell / IFC parser	Geometry extraction and semantic anchors	≥ 95% import success
XR	Unity, OpenXR, XR Interaction Toolkit	Instantiation, physics, interaction, in-scene assessment	≥ 72 FPS (90 FPS target)
Interface	Versioned JSON/REST CS-ISR contract	AI ↔ Unity data exchange	≥ 99% schema parse success

Prototype demonstration - Initial scope contains one unfinished-building floor with two hazard families: 1.fall hazards and 2.struck-by/material-handling hazards. Example prompt: "Create a novice material-transport task with one subtle fall hazard and one blocked pedestrian route." Target: maximum 5 of distinct, valid variants generated from a single learning goal, giving a small working demonstration of the pipeline rather than a claim of full deployment.

8. Methodology and Evaluation Plan

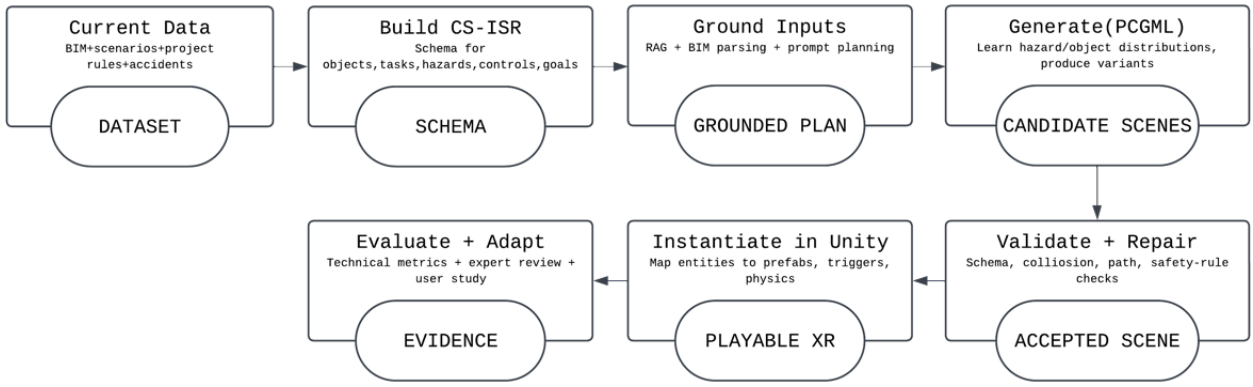


Figure 3. Seven-step methodology from data selection through evaluation and closed-loop adaptation.

- ***Technical test -*** Generate 100–200 scenarios then compare LLM-only planning, rule-based procedural generation, and the full PCGML + validation pipeline.
- ***Ablation -*** To examine the effect of the validation layers, PCGML will first be tested on its own, then with geometry constraints, and finally with both geometry constraints and safety rules. This will show how each additional layer affects the validity and diversity of generated scenarios, how often automatic repair is needed, and how much additional processing time it introduces.
- ***Expert study -*** Construction-safety/BIM reviewers score 20–30 stratified scenarios for realism, hazard correctness, task plausibility, and educational relevance; inter-rater agreement is reported.
- ***Training pilot -*** Static/manual XR vs. adaptive generated XR using identical learning objectives. Primary outcomes: hazard-recognition accuracy and time-to-detection; secondary: unsafe actions, retention, usability, and perceived realism. As this pilot involves human participants, it will proceed only after IRB/ethics approval and informed consent, following institutional research-ethics requirements.

Table 3 demonstrates how the system will be evaluated, with measurement units. The listed targets are provisional, informed by related PCGML and BIM-safety-checking work rather than derived from our own pilot data; actual results will differ once tested, and targets will be revised once early prototype and pilot data are available.

Dimension	Metric	Target	Purpose
Generation speed	Time to generate one scenario	≤ 10 s	Interactive authoring
Content validity	Schema-valid & safety-rule-compliant outputs	≥ 95 of 100 scenes	Reliability & safety grounding
Physical plausibility	Collision-free, reachable scenes	≥ 98 of 100 scenes	Playable scenarios
Authoring efficiency	Time to build one training scenario	~6 hrs (manual) → ≤ 2 hrs (AI-assisted)	Practical value
Training impact	Hazard-detection time & post-test accuracy	≥ 3 s faster; +1–2 hazards correctly identified (out of 10)	Learning effectiveness

9. Expected Contributions and Impact

Scientific - A formal CS-ISR and a safety-constrained PCGML formulation that connects construction information, generative modeling, and adaptive immersive training.

Technical - A working Unity XR pipeline that converts grounded knowledge into validated, playable scenarios and closes the loop using trainee performance data.

Construction safety - Faster developing of site-specific training, greater hazard variation across trades, and repeated, targeted practice on the hazards a given worker actually misses.

10. Scope and Risk Control

Limited scope - The primary study incorporates a handful of construction principles and two categories of risks. (1) enough to substantiate the original claim without presuming completely free of any-restrictions generation, (2) comprehensive regulatory compliance.

Risk control - The outputs of the LLMs cannot be accepted as a source of truth when it comes to safety. The provenance of the content retrieved, schema validation, verification by means of geometry and safety checks, and expert review function as separate barriers before any scenario being incorporated in the training process.

References

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